

1. Agriculture and Farmer Welfare in India

Introduction

Agriculture continues to be one of the most important sectors of the Indian economy. A large part of India's population depends directly or indirectly on agriculture for employment and livelihood.

Government policies in July 2026 continued to focus on:

- Improving farmer income
- Increasing agricultural productivity
- Providing financial support
- Promoting modern farming techniques
- Strengthening rural development

Agricultural development is important not only for food security but also for economic growth and rural stability.

Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)

Introduction

The Pradhan Mantri Kisan Samman Nidhi (PM-KISAN) is a major central government scheme designed to provide financial assistance to eligible farmers.

The scheme provides direct income support to farmer families through Direct Benefit Transfer (DBT).

Background

PM-KISAN was launched in February 2019 by the Government of India.

The scheme was introduced to provide financial assistance to small and marginal farmers who often face challenges such as:

- Rising agricultural costs
- Uncertain crop prices
- Dependence on seasonal income
- Natural disasters affecting crops

Objectives of PM-KISAN

1. Financial Support to Farmers

The scheme provides direct financial assistance to eligible farmers to support agricultural and household requirements.

2. Support Agricultural Activities

Farmers can use the financial assistance for:

- Purchase of seeds
- Fertilisers
- Agricultural equipment
- Farming expenses

3. Digital Transfer of Benefits

The scheme uses technology-based systems for transparent transfer of funds.

Important features:

- Aadhaar-based verification

- Bank account transfer
- Digital monitoring

Importance of PM-KISAN

Economic Importance

The scheme helps improve rural purchasing power and supports the rural economy.

When farmers receive financial support:

- Agricultural spending increases.
- Local markets benefit.
- Rural economic activity improves.

Social Importance

The scheme provides income security to farmers and reduces financial pressure during agricultural seasons.

Important Points

- Scheme: Pradhan Mantri Kisan Samman Nidhi
- Launched: 2019
- Ministry: Ministry of Agriculture and Farmers Welfare
- Sector: Agriculture
- Benefit delivery: Direct Benefit Transfer

2. India Semiconductor Mission and Semiconductor Development

Introduction

Semiconductors are among the most important technologies in the modern world. They are the foundation of digital technology and are used in almost every electronic device.

Modern industries such as:

- Artificial Intelligence
- Defence
- Space technology
- Electric vehicles
- Telecommunications

depend on semiconductor chips.

What are Semiconductors?

A semiconductor is a material that has electrical properties between a conductor and an insulator.

The most commonly used semiconductor material is:

Silicon

Uses of Semiconductor Chips

Electronics

Semiconductor chips are used in:

- Smartphones
- Computers
- Tablets
- Smart devices

Automotive Industry

Modern vehicles use semiconductor chips for:

- Electric vehicle systems
- Safety systems
- Navigation
- Automation

Defence Sector

Semiconductors are used in:

- Radar systems
- Communication systems
- Missile technology
- Surveillance equipment

Space Technology

Space organisations use semiconductor-based systems in:

- Satellites
- Space instruments
- Communication equipment

India Semiconductor Mission (ISM)

Background

India launched the India Semiconductor Mission in 2021 to develop semiconductor manufacturing capability within the country.

The mission works under:

Ministry of Electronics and Information Technology (MeitY)

Objectives of India Semiconductor Mission

1. Semiconductor Manufacturing

India aims to develop:

- Semiconductor fabrication plants
- Chip manufacturing units
- Packaging and testing facilities

2. Reduce Import Dependency

Currently, many semiconductor components are imported from other countries.

Domestic manufacturing helps India:

- Improve technology security
- Reduce supply chain risks
- Strengthen industries

3. Develop Skilled Workforce

The semiconductor industry requires experts in:

- Electronics engineering
- Chip design
- Manufacturing technology
- Research and development

Importance for India

Economic Impact

The semiconductor industry can:

- Create high-value employment
- Increase electronics production
- Support exports
- Attract global companies

Strategic Importance

Semiconductors are essential for national security because they are used in:

- Defence systems
- Cyber infrastructure
- Communication networks

Important Points

- Mission: India Semiconductor Mission
- Started: 2021
- Ministry: MeitY
- Main material: Silicon
- Sector: Electronics and Technology

3. Artificial Intelligence Development in India

Introduction

Artificial Intelligence (AI) has become one of the most important technologies shaping the modern economy.

AI enables computers and machines to perform tasks that normally require human intelligence.

Examples:

- Learning
- Decision making
- Problem solving
- Language understanding

Growth of AI in India

India is focusing on AI development in different sectors:

- Healthcare
- Agriculture
- Education
- Defence
- Governance

Applications of Artificial Intelligence

Agriculture

AI helps farmers through:

- Weather prediction
- Crop monitoring

- Disease detection
- Smart irrigation

This improves productivity and reduces losses.

Healthcare

AI is used for:

- Disease diagnosis
- Medical image analysis
- Drug research
- Healthcare management

Education

AI supports:

- Personalised learning
- Digital education platforms
- Automated assessment systems

Defence

AI applications include:

- Surveillance systems
- Cybersecurity
- Autonomous systems
- Data analysis

Challenges of AI

Important challenges include:

Data Privacy

Protection of personal information is necessary.

Cybersecurity

AI systems can become targets of cyber attacks.

Ethical Issues

Responsible use of AI is required.

Importance of AI for India

AI can help India in:

- Economic growth
- Innovation
- Digital transformation
- Global technology competition

Important Points

- Technology: Artificial Intelligence
- Applications: Healthcare, Agriculture, Defence, Education
- Major concerns: Privacy and Security

4. Space Technology and ISRO Developments

Introduction

India's space programme plays an important role in communication, weather forecasting, navigation, disaster management and scientific research.

The Indian Space Research Organisation (ISRO) is responsible for India's space activities.

Indian Space Research Organisation (ISRO)

Background

ISRO was established in 1969 with the aim of developing India's space capability.

Headquarters

Bengaluru, Karnataka

Founder

Dr. Vikram Sarabhai is considered the father of India's space programme.

Major Areas of ISRO Activities

1. Communication Satellites

Used for:

- Television broadcasting
- Internet services
- Communication networks

2. Earth Observation Satellites

Used for:

- Agriculture monitoring
- Disaster management
- Weather studies
- Environmental monitoring

3. Navigation Technology

India developed:

NavIC (Navigation with Indian Constellation)

It provides regional navigation services.

Important Space Missions

Chandrayaan Programme

Purpose:

Moon exploration and scientific study.

Aditya-L1 Mission

Purpose:

Study the Sun and understand solar activities.

Gaganyaan Mission

Purpose:

India's human spaceflight programme.

Importance of Space Technology

Economic Benefits

- Supports communication services
- Encourages space startups
- Creates technology development

Security Benefits

Space technology supports:

- Defence communication
- Surveillance
- Navigation

Important Points

- Organisation: ISRO
- Established: 1969
- Headquarters: Bengaluru
- Founder: Vikram Sarabhai

5. Renewable Energy Development in India

Introduction

India is focusing on renewable energy to meet increasing energy demand and reduce dependence on fossil fuels.

Renewable energy is important for:

- Energy security
- Environmental protection
- Sustainable development

Major Renewable Energy Sources

Solar Energy

Solar power is one of India's fastest-growing renewable energy sectors.

Uses:

- Solar power plants
- Rooftop solar systems

Wind Energy

Wind energy contributes significantly to India's clean energy capacity.

Major states:

- Tamil Nadu
- Gujarat
- Karnataka
- Maharashtra

Green Hydrogen

Green hydrogen is produced using renewable energy.

Applications:

- Clean transportation
- Industry
- Energy storage

International Solar Alliance (ISA)

Background

The International Solar Alliance was launched to promote solar energy cooperation among countries.

Headquarters

Gurugram, Haryana

Founded

2015

Importance

ISA promotes:

- Solar energy development
- Clean energy investment
- International cooperation

Importance of Renewable Energy for India

Renewable energy helps:

- Reduce carbon emissions
- Improve energy independence
- Support climate commitments
- Create green jobs

SECTION 1: INDIAN ECONOMY

1. Indian Economic Growth and Development

Introduction

India continued its journey as one of the world's major emerging economies. Economic development in India depends on multiple sectors including:

- Agriculture
- Manufacturing
- Services
- Digital economy
- Infrastructure development

The government and financial institutions continued focusing on maintaining economic growth while managing challenges such as inflation, global economic uncertainty and employment generation.

Major Factors Affecting Indian Economy

1. Domestic Demand

India's large population creates strong demand for:

- Consumer goods
- Housing
- Transportation
- Digital services

Domestic consumption remains an important driver of economic growth.

2. Infrastructure Development

Infrastructure investment supports economic growth through:

- Roads
- Railways
- Airports
- Ports
- Digital networks

Better infrastructure improves connectivity and business activities.

3. Manufacturing Growth

India is promoting manufacturing through initiatives such as:

- Make in India
- Production Linked Incentive (PLI) schemes

The objective is to increase domestic production and attract investment.

4. Digital Economy

India's digital economy continues to expand through:

- Digital payments
- Online services
- Technology-based businesses

Digital transformation has increased financial inclusion and improved service delivery.

Important Economic Concepts

Gross Domestic Product (GDP)

GDP represents the total value of goods and services produced within a country during a specific period.

GDP is an important indicator of economic performance.

Inflation

Inflation refers to a continuous increase in the prices of goods and services.

High inflation affects:

- Purchasing power
- Savings
- Household expenses

Fiscal Deficit

Fiscal deficit occurs when government expenditure is higher than government revenue.

It indicates the borrowing requirement of the government.

Importance of Economic Growth

Economic growth helps in:

- Employment generation
- Poverty reduction
- Infrastructure development
- Improvement in living standards

Important Points

- Economy depends on agriculture, industry and services.
- Digital economy is becoming a major growth area.
- Infrastructure development supports long-term growth.
- Inflation control is important for economic stability.

2. Reserve Bank of India (RBI) Monetary Policy

Introduction

The Reserve Bank of India is the central banking institution of India.

It controls monetary policy and plays an important role in maintaining:

- Price stability
- Banking system stability
- Economic growth

Reserve Bank of India (RBI)

Background

The RBI was established in 1935 under the Reserve Bank of India Act, 1934.

It was nationalised in 1949.

Headquarters

Mumbai, Maharashtra

Main Functions of RBI

1. Monetary Policy Management

RBI controls money supply and interest rates to manage inflation and economic growth.

2. Regulation of Banks

RBI supervises:

- Commercial banks
- Cooperative banks
- Financial institutions

It ensures stability of the banking system.

3. Currency Management

RBI is responsible for:

- Issuing currency notes
- Managing currency circulation

Note:

The ₹1 note is issued by the Government of India.

4. Foreign Exchange Management

RBI manages foreign exchange reserves and regulates foreign currency transactions.

Monetary Policy Tools

Repo Rate

Repo rate is the interest rate at which RBI lends money to commercial banks.

Effect of Increase:

- Loans become expensive
- Borrowing decreases
- Inflation may reduce

Reverse Repo Rate

The rate at which RBI borrows money from banks.

It helps control excess liquidity in the banking system.

Cash Reserve Ratio (CRR)

CRR is the percentage of deposits banks must maintain with RBI as cash reserves.

Purpose:

- Control credit creation
- Maintain financial stability

Statutory Liquidity Ratio (SLR)

Banks maintain a portion of deposits in liquid assets such as government securities.

Importance of RBI

RBI decisions affect:

- Home loans
- Business loans
- Investment
- Inflation
- Economic activity

Important Points

- Established: 1935
- Nationalised: 1949
- Headquarters: Mumbai
- Function: Central Banking Institution

3. Banking Sector Development and Financial Inclusion

Introduction

India's banking sector plays a major role in economic development by providing:

- Savings facilities
- Loans
- Payment services
- Financial support

Financial Inclusion

Meaning

Financial inclusion means providing affordable financial services to all sections of society.

It focuses on:

- Rural population
- Low-income groups
- Small businesses

Major Areas of Financial Inclusion

1. Bank Accounts

Bank accounts help citizens:

- Save money
- Receive government benefits
- Access financial services

2. Digital Banking

Digital banking allows customers to access services through:

- Mobile applications
- Internet banking
- Digital platforms

3. Digital Payments

India's digital payment system has expanded through:

- UPI
- Mobile payments
- Online transactions

Unified Payments Interface (UPI)

Introduction

UPI is an instant digital payment system that allows money transfer between bank accounts.

Developed By

National Payments Corporation of India (NPCI)

Importance

UPI has improved:

- Digital payments
- Financial accessibility
- Small business transactions

Important Points

- UPI is a major digital payment innovation.
- NPCI developed UPI.

- Digital payments support financial inclusion.

SECTION 2: DEFENCE AND SECURITY

4. Defence Modernisation in India

Introduction

India continues modernising its defence capabilities to strengthen national security.

Modern defence requires advanced technology including:

- Artificial intelligence
- Cybersecurity
- Space technology
- Unmanned systems

Objectives of Defence Modernisation

1. Strengthening Military Capability

India focuses on improving:

- Army equipment
- Naval systems
- Air force technology

2. Reducing Defence Imports

Through indigenous manufacturing, India aims to reduce dependence on foreign defence equipment.

3. Promoting Defence Industry

India encourages:

- Defence startups
- Private companies
- Research organisations

Atmanirbhar Bharat in Defence

Introduction

The initiative promotes domestic production of defence equipment.

Main Focus Areas

Aircraft Manufacturing

Development of:

- Fighter aircraft
- Helicopters
- Aerospace systems

Missile Technology

Development of:

- Missile systems
- Defence technology

Naval Development

Focus on:

- Warships
- Submarines
- Maritime security systems

Importance

Defence self-reliance helps:

- National security
- Technology development
- Employment generation

Important Defence Organisations

Defence Research and Development Organisation (DRDO)

Introduction

DRDO is India's premier defence research organisation.

Established

1958

Headquarters

New Delhi

Functions

Develops:

- Missiles
- Defence equipment
- Military technology

Hindustan Aeronautics Limited (HAL)

Introduction

HAL is an Indian aerospace and defence company.

Headquarters

Bengaluru, Karnataka

Functions

Manufactures:

- Aircraft
- Helicopters
- Aerospace equipment

5. Border Security and National Security

Introduction

India's security policy focuses on protecting its land borders, maritime areas and cyberspace.

National security includes:

- Military security
- Cyber security
- Border management

- Internal security

Border Security Areas

Land Borders

India shares borders with:

- China
- Pakistan
- Nepal
- Bhutan
- Bangladesh
- Myanmar

Maritime Security

India focuses on protecting:

- Indian Ocean region
- Coastal areas
- Maritime trade routes

Role of Armed Forces

Indian Army

Responsible for:

- Land-based defence
- Border protection

Indian Navy

Responsible for:

- Maritime security
- Naval operations

Indian Air Force

Responsible for:

- Air defence
- Aerial operations

Importance of Maritime Security

The Indian Ocean is strategically important because:

- Major trade routes pass through it.
- It connects Asia, Africa and Europe.

Important Points

- National security includes military and non-military threats.
- Maritime security is important for trade protection.
- Modern warfare requires advanced technology.

6. Cyber Security and Digital Threat Management

Introduction

With rapid digital growth, cybersecurity has become an important part of national security.

Cyber threats can affect:

- Government systems
- Banking networks
- Critical infrastructure
- Personal data

Major Cyber Threats

1. Cyber Fraud

Includes:

- Online financial scams
- Identity theft

2. Data Theft

Unauthorised access to:

- Personal information
- Government data

3. Cyber Attacks on Infrastructure

Targets include:

- Power systems
- Communication networks
- Financial institutions

CERT-In

Full Form

Indian Computer Emergency Response Team

Role

CERT-In works for:

- Cyber threat monitoring
- Security alerts
- Incident response

Importance of Cyber Security

Cybersecurity protects:

- Digital economy
- National infrastructure
- Citizens' data

SECTION 1: SCIENCE & TECHNOLOGY

1. Artificial Intelligence (AI) Development in India

Introduction

Artificial Intelligence (AI) has become one of the most important technologies in the 21st century. AI enables machines and computer systems to perform tasks that normally require human intelligence.

These tasks include:

- Learning from data
- Recognising patterns
- Making decisions
- Understanding languages
- Solving complex problems

India is focusing on Artificial Intelligence to improve economic growth, public services, scientific research and national security.

Growth of Artificial Intelligence in India

India has a large technology workforce and a growing digital economy, making it an important country in AI development.

AI applications are increasing in areas such as:

- Healthcare
- Agriculture

- Education
- Defence
- Banking
- Governance

Applications of Artificial Intelligence

1. AI in Agriculture

Agriculture is one of the major sectors where AI can improve productivity.

AI technology helps farmers through:

Weather Prediction

AI-based systems analyse weather patterns and provide information about:

- Rainfall
- Temperature
- Climate conditions

This helps farmers plan agricultural activities.

Crop Monitoring

AI can analyse satellite images and agricultural data to monitor:

- Crop health
- Growth patterns
- Disease conditions

Smart Farming

AI supports:

- Automated irrigation systems
- Soil analysis
- Better resource management

2. AI in Healthcare

AI is transforming healthcare by improving diagnosis and treatment.

Applications include:

Disease Detection

AI systems can analyse medical images to identify diseases.

Examples:

- Cancer detection
- Eye disease identification
- Heart-related conditions

Medical Research

AI helps researchers in:

- Drug discovery
- Vaccine development
- Disease prediction

Digital Healthcare

AI supports:

- Online consultation
- Patient data management
- Healthcare planning

3. AI in Defence

Modern defence systems increasingly use AI technology.

Applications include:

- Surveillance systems
- Autonomous vehicles
- Cybersecurity
- Military data analysis

4. AI in Governance

Government departments use AI for:

- Data analysis
- Improving public services
- Detecting fraud
- Decision-making support

National Strategy on Artificial Intelligence

Background

India has been promoting AI development through government initiatives and research programmes.

NITI Aayog released a national discussion paper on AI development.

Important Areas of AI Development

- AI research
- Skill development
- Data infrastructure
- Ethical AI usage
- Innovation ecosystem

Challenges of Artificial Intelligence

Data Privacy

AI systems require large amounts of data. Protecting personal information is important.

Cyber Security

AI systems can become targets of cyber attacks.

Employment Impact

Automation may change traditional job roles and requires new skills.

Importance of AI for India

AI can help India in:

- Economic development
- Scientific innovation

- Better governance
- Global technology competition

Important Points

- AI enables machines to perform intelligent tasks.
- Major application areas are healthcare, agriculture, defence and governance.
- Data security and ethical use are important challenges.

2. Semiconductor Industry Development in India

Introduction

Semiconductors are essential components of modern electronic systems.

Almost every advanced technology depends on semiconductor chips.

They are used in:

- Mobile phones
- Computers
- Electric vehicles
- Satellites
- Defence equipment
- Artificial Intelligence systems

What are Semiconductors?

Semiconductors are materials that have electrical properties between conductors and insulators.

The most commonly used semiconductor material is:

Silicon

Importance of Semiconductor Chips

Digital Technology

Semiconductor chips are the foundation of:

- Smartphones
- Computers
- Digital devices

Automotive Sector

Modern vehicles use chips for:

- Electric vehicle systems
- Safety features
- Navigation systems

Defence Sector

Military systems depend on advanced chips for:

- Communication
- Radar
- Missile systems

India Semiconductor Mission

Background

India launched the India Semiconductor Mission to develop domestic semiconductor manufacturing capability.

Objectives

1. Semiconductor Manufacturing

Develop:

- Chip fabrication units
- Testing facilities
- Packaging industries

2. Reduce Import Dependency

India currently imports many semiconductor components.

Domestic manufacturing helps:

- Improve supply security
- Reduce external dependency

3. Develop Skilled Workforce

The semiconductor sector requires experts in:

- Electronics
- Engineering
- Chip design
- Research

Impact on India

Economic Impact

- Creates skilled employment
- Increases electronics production
- Attracts investment

Strategic Impact

Semiconductor technology is important for:

- Defence
- Space
- Digital infrastructure

Important Points

- Semiconductor material: Silicon
- Sector: Electronics and Technology
- Important for AI, defence and communication systems

SECTION 2: SPACE TECHNOLOGY

3. Indian Space Programme and ISRO Development

Introduction

India's space programme plays an important role in communication, navigation, weather forecasting and scientific research.

The Indian Space Research Organisation (ISRO) is responsible for India's space activities.

Indian Space Research Organisation (ISRO)

Establishment

1969

Headquarters

Bengaluru, Karnataka

Founder

Dr. Vikram Sarabhai

Major Activities of ISRO

1. Satellite Technology

Satellites are used for:

- Communication
- Weather monitoring
- Earth observation
- Navigation

2. Earth Observation

Earth observation satellites help in:

- Agriculture monitoring
- Disaster management
- Environmental studies

3. Navigation System – NavIC

Full Form:

Navigation with Indian Constellation

Purpose:

Provides navigation services using Indian satellites.

Applications:

- Transportation
- Disaster management
- Defence

Major Indian Space Missions

Chandrayaan Programme

Purpose:

Moon exploration and scientific research.

Importance:

Helps scientists study:

- Lunar surface
- Moon minerals
- Space environment

Aditya-L1 Mission

Purpose:

Study the Sun.

Studies:

- Solar atmosphere
- Solar storms
- Solar activity

Gaganyaan Mission

Purpose:

India's human spaceflight programme.

Objective:

Send Indian astronauts into space.

Importance of Space Technology

Economic Importance

Supports:

- Communication services
- Space startups
- Technology development

Security Importance

Supports:

- Defence communication
- Surveillance
- Navigation

Important Points

- ISRO established in 1969.
- Headquarters: Bengaluru.
- NavIC is India's navigation system.

SECTION 3: HEALTH AND BIOTECHNOLOGY

4. Digital Health Development in India

Introduction

Digital technology is transforming healthcare delivery in India.

Digital health aims to improve:

- Healthcare accessibility
- Medical record management
- Patient services

Major Areas of Digital Health

1. Telemedicine

Telemedicine allows doctors and patients to communicate through digital platforms.

Benefits:

- Healthcare access in rural areas
- Reduced travel requirement

2. Digital Health Records

Electronic health records help maintain:

- Patient history
- Medical information
- Treatment details

3. Health Data Management

Digital systems help government agencies monitor:

- Diseases
- Healthcare requirements
- Public health trends

Biotechnology Development

Introduction

Biotechnology combines biology and technology to develop solutions in healthcare, agriculture and industry.

Applications of Biotechnology

Healthcare

Used in:

- Vaccines
- Medicines
- Genetic research

Agriculture

Used for:

- Improved crops
- Disease resistance
- Better productivity

Importance of Biotechnology

Biotechnology supports:

- Medical innovation
- Food security
- Scientific development

Important Points

- Digital health improves healthcare accessibility.
- Biotechnology supports medicine and agriculture.
- Technology plays an important role in healthcare development.

SECTION 4: ENVIRONMENT AND CLIMATE

5. Renewable Energy Development in India

Introduction

India is focusing on renewable energy to meet increasing energy demand while reducing environmental impact.

Renewable energy sources are naturally available and can be replenished.

Major Renewable Energy Sources

Solar Energy

Solar energy uses sunlight to generate electricity.

Applications:

- Solar power plants
- Rooftop solar systems

Wind Energy

Wind energy converts wind power into electricity.

Major states:

- Tamil Nadu

- Gujarat
- Karnataka
- Maharashtra

Green Hydrogen

Green hydrogen is produced using renewable energy.

Uses:

- Industry
- Transport
- Energy storage

Importance of Renewable Energy

Energy Security

Reduces dependence on imported fossil fuels.

Environmental Benefits

Helps reduce:

- Carbon emissions
- Air pollution

Economic Benefits

Creates:

- Green jobs

- New industries
- Investment opportunities

International Solar Alliance (ISA)

Background

The International Solar Alliance promotes cooperation among countries for solar energy development.

Headquarters

Gurugram, Haryana

Founded

2015

Initiative

India and France

Important Points

- Renewable energy supports sustainable development.
- Solar energy is a major focus area.
- Green hydrogen is an emerging clean energy technology.

SECTION 5: IMPORTANT GOVERNMENT INITIATIVES

6. Digital India Programme

Introduction

Digital India is a major government initiative aimed at transforming India into a digitally empowered society.

Launched

2015

Objectives

Digital Infrastructure

Develop:

- Internet connectivity
- Digital platforms
- Technology systems

Digital Services

Provides:

- Online government services
- Digital documents
- Citizen services

Digital Payments

Promotes:

- Cashless transactions
- Financial inclusion

Importance

Digital India improves:

- Transparency
- Efficiency
- Public service delivery

7. Startup India Initiative

Introduction

Startup India promotes entrepreneurship and innovation in India.

Launched

2016

Objectives

Encourage Entrepreneurship

Supports individuals starting new businesses.

Promote Innovation

Focus areas:

- Technology
- Manufacturing
- Biotechnology
- Artificial Intelligence

Employment Generation

Startups create:

- New jobs
- New economic opportunities

Importance

Startup ecosystem helps India develop:

- Innovation
- Technology solutions
- Business growth

SECTION 1: GOVERNMENT SCHEMES AND INITIATIVES

1. Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)

Introduction

Agriculture remains one of the most important sectors of India's economy. A large number of Indian families depend on farming and related activities for their livelihood.

To provide financial support to farmers, the Government of India introduced the **Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)** scheme.

The scheme provides direct income support to eligible farmer families through the Direct Benefit Transfer (DBT) system.

Background

PM-KISAN was launched in **2019**.

The main purpose of the scheme was to provide financial assistance to small and marginal farmers who face challenges such as:

- Increasing agricultural costs
- Uncertain crop income

- Natural disasters
- Market price fluctuations

Objectives of PM-KISAN

1. Providing Income Support

The scheme provides financial assistance directly to eligible farmers.

This helps farmers manage:

- Agricultural expenses
- Household requirements
- Farming investments

2. Supporting Agricultural Activities

Farmers can use financial support for:

- Seeds
- Fertilisers
- Farming equipment
- Crop-related expenses

3. Promoting Digital Transparency

The scheme uses:

- Aadhaar verification
- Bank account transfers
- Digital monitoring systems

This reduces delays and improves transparency.

Importance of PM-KISAN

Economic Impact

The scheme supports rural purchasing power.

Benefits:

- Improves rural economy
- Supports agricultural spending
- Helps small farmers

Social Impact

The scheme provides financial stability to farming communities.

Important Points

- Full Form: Pradhan Mantri Kisan Samman Nidhi
- Launched: 2019
- Ministry: Ministry of Agriculture and Farmers Welfare
- Type: Central Sector Scheme
- Benefit Transfer: Direct Benefit Transfer

2. Pradhan Mantri Surya Ghar Yojana

Introduction

India is focusing on renewable energy to reduce dependence on fossil fuels and increase clean energy production.

The **Pradhan Mantri Surya Ghar: Muft Bijli Yojana** is an important government initiative promoting rooftop solar energy.

Background

The scheme was launched in 2024.

Its objective is to encourage households to install rooftop solar systems and generate clean electricity.

Objectives

1. Promote Solar Energy

The scheme encourages households to adopt solar power.

2. Reduce Electricity Costs

Solar power generation can help households reduce electricity expenses.

3. Increase Renewable Energy Capacity

The scheme supports India's clean energy goals.

Benefits

For Citizens

- Lower electricity bills
- Clean energy access
- Energy independence

For Environment

- Reduced carbon emissions
- Less dependence on fossil fuels

Importance

The scheme supports:

- Renewable energy development
- Climate action
- Sustainable growth

Important Points

- Scheme: PM Surya Ghar: Muft Bijli Yojana
- Sector: Renewable Energy

- Focus: Rooftop Solar Power

3. Ayushman Bharat Digital Mission (ABDM)

Introduction

Digital technology is transforming healthcare services in India.

The Ayushman Bharat Digital Mission aims to create a digital health ecosystem where healthcare information can be managed securely.

Background

The mission was launched in 2021.

It is implemented by the Government of India to improve digital healthcare services.

Objectives

1. Digital Health Records

The mission promotes electronic health records for citizens.

2. Healthcare Accessibility

Digital platforms help people access healthcare services easily.

3. Connecting Healthcare Providers

The system connects:

- Hospitals
- Doctors
- Patients
- Health service providers

Importance

Digital healthcare improves:

- Healthcare delivery
- Medical information management
- Patient convenience

Important Points

- Mission: Ayushman Bharat Digital Mission
- Sector: Healthcare
- Focus: Digital health ecosystem

SECTION 2: GOVERNMENT POLICIES AND REFORMS

4. National Education Policy (NEP)

Implementation

Introduction

Education plays a major role in national development. India introduced the National Education Policy 2020 to transform the education system.

Objectives of NEP 2020

1. Holistic Education

The policy focuses on:

- Knowledge development
- Skills
- Creativity
- Critical thinking

2. Skill-Based Learning

Education is moving towards practical skills and vocational training.

3. Technology Integration

Digital tools are promoted for:

- Online learning
- Educational resources
- Better access

Major Features

New School Structure

The policy introduced:

5+3+3+4 education structure

It replaced the earlier 10+2 structure.

Mother Tongue-Based Learning

The policy encourages education in regional languages during early education stages.

Importance

NEP aims to create:

- Skilled workforce
- Innovation capacity
- Better learning outcomes

Important Points

- Policy: National Education Policy
- Year: 2020
- Focus: Education reform and skill development

5. Make in India and Production Linked Incentive (PLI) Scheme

Introduction

India is promoting manufacturing growth through policies that encourage domestic production and investment.

Make in India

Launched

2014

Objectives

- Increase manufacturing
- Attract foreign investment
- Create employment opportunities

Production Linked Incentive (PLI)

Introduction

PLI schemes provide financial incentives to companies based on production performance.

Main Sectors Covered

PLI supports industries including:

- Electronics
- Pharmaceuticals
- Automobile
- Solar equipment
- Telecommunications

Importance

PLI helps:

- Increase manufacturing capacity
- Improve exports
- Create jobs
- Reduce imports

Important Points

- Make in India launched: 2014
- PLI promotes manufacturing growth
- Focus: Domestic production and exports

SECTION 3: IMPORTANT REPORTS AND INDEXES

6. NITI Aayog Reports and Development Indicators

Introduction

NITI Aayog is India's policy think tank that supports government planning and development strategies.

Background

NITI Aayog was established in:

2015

It replaced the Planning Commission.

Functions

Policy Research

Studies development challenges and suggests solutions.

Monitoring Development

Tracks progress in areas such as:

- Health
- Education
- Poverty reduction

- Sustainable development

Cooperative Federalism

Encourages cooperation between:

- Central Government
- State Governments

Important Reports by NITI Aayog

Examples:

- Sustainable Development Goals Index
- Innovation Index
- Health Index

Importance

Reports help:

- Measure development
- Identify challenges
- Improve policymaking

Important Points

- Organisation: NITI Aayog
- Established: 2015
- Replaced: Planning Commission

7. Reserve Bank of India Reports

Introduction

The Reserve Bank of India publishes various reports related to India's financial and economic system.

Important RBI Reports

Financial Stability Report

It analyses:

- Banking sector health
- Financial risks
- Economic stability

Monetary Policy Reports

They discuss:

- Inflation trends
- Interest rates
- Economic conditions

Importance

RBI reports help understand:

- Banking performance
- Financial risks
- Economic trends

SECTION 4: IMPORTANT APPOINTMENTS

8. Constitutional and Institutional Appointments

Introduction

Appointments to important government institutions play a major role in administration and policymaking.

Important appointments are made in areas including:

- Judiciary
- Defence
- Finance
- Administration
- Regulatory bodies

Importance of Appointments

Leadership positions influence:

- Policy implementation
- Institutional functioning
- Decision-making

Major Areas of Appointments

Financial Institutions

Includes:

- Banking regulators
- Economic institutions

Defence Organisations

Includes:

- Military leadership
- Defence administration

Government Bodies

Includes:

- Constitutional authorities
- Administrative organisations

SECTION 5: AWARDS AND HONOURS

9. National Awards and Recognition

Introduction

Awards recognise achievements in different fields including:

- Science
- Technology
- Sports
- Literature
- Public service

Major Award Categories

Science and Technology Awards

Recognise:

- Research achievements
- Innovation
- Scientific contributions

Sports Awards

Recognise:

- Athlete achievements
- Coaching excellence

Civil Awards

Recognise contributions to:

- Society
- Public service
- National development

Importance

Awards encourage:

- Innovation
- Excellence
- Social contribution

SECTION 6: IMPORTANT EVENTS

10. Infrastructure and Development Activities

Introduction

Infrastructure development is essential for India's economic growth.

Government continues focusing on:

- Transport infrastructure
- Digital infrastructure
- Rural development
- Urban development

Major Infrastructure Areas

Roads and Highways

Importance:

- Better connectivity
- Trade improvement
- Economic activity

Railways

Focus areas:

- Modernisation
- Safety improvement
- Faster transportation

Digital Infrastructure

Includes:

- Internet connectivity
- Digital services
- Technology platforms

Importance

Infrastructure development helps:

- Increase investment
- Create employment
- Improve connectivity

- Support economic growth

SECTION 1: INTERNATIONAL RELATIONS

1. Changing Global Order and Multipolar World

Introduction

The global political system is undergoing major changes. Earlier, international politics was largely influenced by a few powerful countries. However, emerging economies and regional organisations are now playing a larger role.

The world is moving towards a **multipolar system**, where several countries and groups influence global decisions.

Meaning of Multipolar World

A multipolar world means:

- Multiple countries have significant political, economic and military influence.
- International decisions are shaped by cooperation and competition among different powers.

Major Global Power Centres

United States

The United States continues to have major influence through:

- Military capability
- Global financial system
- Advanced technology
- International alliances

China

China plays an important role through:

- Manufacturing power
- Trade networks
- Infrastructure investment
- Military development

European Union

The European Union influences:

- International trade rules
- Climate policies
- Economic regulations

India

India's global role is increasing because of:

- Large economy
- Technology sector
- Defence capability
- Leadership among developing nations

Factors Driving Global Changes

1. Economic Competition

Countries compete for:

- Trade opportunities
- Technology leadership
- Energy resources

2. Technology Competition

Important areas include:

- Artificial Intelligence
- Semiconductor technology
- Cybersecurity
- Space technology

3. Security Challenges

Global security issues include:

- Regional conflicts
- Terrorism
- Maritime security
- Cyber threats

India's Position

India follows the principle of **strategic autonomy**.

This means India:

- Cooperates with different countries.
- Makes independent foreign policy decisions.
- Balances relations with multiple global powers.

Important Points

- World order is becoming multipolar.
- Emerging economies are gaining importance.
- Technology and security are major factors shaping international relations.

SECTION 2: UNITED NATIONS AND GLOBAL GOVERNANCE

2. United Nations Sustainable Development Goals (SDGs) Review

Introduction

The United Nations continued reviewing global progress towards achieving the Sustainable Development Goals (SDGs).

The 2026 High-Level Political Forum on Sustainable Development focused on accelerating progress towards the 2030 Agenda.

About Sustainable Development Goals

The SDGs were adopted by the United Nations in:

2015

Target year:

2030

Total Number of SDGs

There are:

17 Sustainable Development Goals

Major SDGs

No Poverty

Objective:

Reduce poverty worldwide.

Zero Hunger

Focus:

- Food security
- Better nutrition
- Sustainable agriculture

Quality Education

Aim:

Provide inclusive and equal education opportunities.

Good Health and Well-being

Focus:

- Healthcare access
- Disease prevention

Climate Action

Aim:

Address climate change and reduce environmental damage.

Challenges in Achieving SDGs

Countries face problems such as:

Climate Change

Extreme weather events affect development.

Economic Inequality

Income differences slow development progress.

Conflicts

Wars and instability affect:

- Food supply
- Healthcare
- Infrastructure

Importance for India

India's development programmes are connected with SDGs through:

- Poverty reduction schemes
- Digital development
- Renewable energy
- Healthcare programmes

Important Points

- SDGs adopted in 2015.
- Target year: 2030.
- There are 17 SDGs.
- UN reviews progress through international forums.

3. United Nations Security Council Reform

Introduction

Reform of the United Nations Security Council (UNSC) has remained an important international issue.

Many countries argue that the present structure does not represent today's global realities.

About United Nations Security Council

The UNSC is responsible for maintaining:

- International peace
- Global security

Permanent Members (P5)

The five permanent members are:

1. United States
2. United Kingdom
3. France
4. Russia
5. China

Powers of Permanent Members

Permanent members have:

Veto Power

They can block important UNSC resolutions.

Need for Reform

Many countries demand reform because:

- Global power distribution has changed.
- Developing countries need greater representation.
- Current structure reflects post-World War II conditions.

India's Position

India supports:

- Permanent membership for India.
- Greater representation for developing nations.
- Reform of global institutions.

Important Points

- UNSC has five permanent members.
- Veto power is held by P5 countries.
- India supports UNSC reform.

SECTION 3: BRICS DEVELOPMENTS

4. BRICS Cooperation and Expansion

Introduction

BRICS is an important grouping of emerging economies that promotes economic cooperation and greater representation of developing countries in global affairs.

BRICS discussions continue to focus on economic cooperation, development finance and climate-related issues.

BRICS Full Form

Originally:

- Brazil
- Russia
- India
- China
- South Africa

Objectives of BRICS

1. Economic Cooperation

BRICS promotes:

- Trade

- Investment
- Economic partnerships

2. Development Finance

BRICS created:

New Development Bank (NDB)

New Development Bank

Headquarters:

Shanghai, China

Purpose:

Provides financial support for:

- Infrastructure projects
- Sustainable development
- Development programmes

3. Reform of Global Institutions

BRICS countries support:

- Greater representation of developing countries.
- Changes in international financial institutions.

Importance of BRICS for India

BRICS provides India:

- International cooperation platform
- Economic opportunities
- Partnership with emerging economies

Important Points

- BRICS represents emerging economies.
- New Development Bank is associated with BRICS.
- BRICS supports greater developing country representation.

SECTION 4: G20 AND GLOBAL ECONOMIC COOPERATION

5. G20 and Global Economic Issues

Introduction

The Group of Twenty (G20) is one of the most important international economic forums.

It brings together major economies to discuss:

- Global economic growth
- Financial stability
- Climate issues

- Development challenges

About G20

Full Form:

Group of Twenty

Members Include

- Major developed economies
- Emerging economies
- European Union
- African Union

Main Areas of Discussion

Global Economic Growth

Countries discuss:

- Economic recovery
- Investment
- Trade

Financial Stability

Focus areas:

- Banking systems
- International finance
- Economic risks

Climate Finance

Countries discuss funding for:

- Clean energy
- Climate adaptation

India's Role in G20

India hosted the G20 Summit in:

2023

Important themes:

- Digital public infrastructure
- Sustainable development
- Global South cooperation

Important Points

- G20 focuses mainly on global economic issues.
- India hosted G20 in 2023.
- It includes developed and developing economies.

SECTION 5: GLOBAL ECONOMY

6. Global Inflation and Economic Challenges

Introduction

The global economy continued facing challenges due to:

- Inflation
- Energy price fluctuations
- Geopolitical tensions
- Supply chain disruptions

Global economic uncertainty remained an important issue in international discussions.

Major Economic Challenges

1. Inflation

Inflation refers to the increase in prices of goods and services.

Effects:

- Reduces purchasing power
- Increases household expenses
- Affects economic growth

2. Energy Security

Energy prices affect:

- Transportation
- Manufacturing
- Inflation levels

Countries focus on:

- Renewable energy
- Stable energy supply

3. Supply Chain Problems

Global supply chains are affected by:

- Conflicts
- Trade restrictions
- Production disruptions

Role of International Institutions

International Monetary Fund (IMF)

Functions:

- Provides financial assistance
- Monitors global economy
- Gives policy recommendations

World Bank

Functions:

- Development financing
- Infrastructure support
- Poverty reduction programmes

Impact on India

Global economic conditions influence:

- Exports
- Foreign investment
- Currency value
- Employment

Important Points

- Inflation affects global economic stability.
- Energy security is a major concern.
- IMF and World Bank support economic development.

SECTION 6: GLOBAL DEFENCE AND SECURITY

7. Indo-Pacific Security Cooperation

Introduction

The Indo-Pacific region has become one of the most strategically important regions in the world.

The region is important because major international trade routes pass through it.

Importance of Indo-Pacific

The region affects:

- Global trade
- Maritime security
- Energy transportation

Major Security Concerns

Maritime Disputes

Countries face challenges related to:

- Territorial claims
- Freedom of navigation

Military Competition

Countries are increasing:

- Naval capability
- Defence technology

QUAD Cooperation

Members:

1. India
2. United States
3. Japan
4. Australia

Areas of Cooperation

- Maritime security
- Disaster management
- Technology cooperation
- Regional stability

Importance for India

Indo-Pacific security is important because:

- India has a long coastline.
- Indian Ocean trade routes are strategically important.

Important Points

- Indo-Pacific is important for global trade.
- QUAD focuses on regional cooperation.
- Maritime security is a major issue.

8. NATO and International Defence Cooperation

Introduction

NATO is one of the world's major military alliances.

NATO

Full Form:

North Atlantic Treaty Organization

Established:

1949

Headquarters:

Brussels, Belgium

Purpose

NATO follows the principle of collective defence.

Meaning:

An attack against one member is considered a concern for all members.

Role of NATO

NATO focuses on:

- Defence cooperation
- Security planning
- Military coordination

Important Points

- NATO was established in 1949.
- Headquarters: Brussels.
- It is a defence alliance.

SECTION 7: CLIMATE AND ENVIRONMENT

9. Global Climate Action and Climate Finance

Introduction

Climate change continues to be one of the biggest global challenges.

Countries are working together to reduce greenhouse gas emissions and protect vulnerable communities.

Major Climate Challenges

Rising Temperatures

Effects:

- Heat waves
- Changing weather patterns
- Environmental damage

Extreme Weather Events

Includes:

- Floods
- Droughts
- Storms

Climate Finance

Developing countries require financial support for:

- Adaptation
- Clean energy transition
- Disaster management

Paris Agreement

Adopted:

2015

Objective:

Limit global temperature rise and reduce greenhouse gas emissions.

Renewable Energy Transition

Countries are promoting:

- Solar energy
- Wind energy
- Green hydrogen
- Electric vehicles

Importance for India

Climate action is important for:

- Agriculture
- Water security
- Energy security
- Sustainable development

Important Points

- Climate change is a global challenge.
- Climate finance is important for developing countries.
- Renewable energy is a key solution.

SECTION 1: SCIENCE AND TECHNOLOGY

1. Global Development of Artificial Intelligence (AI)

Introduction

Artificial Intelligence (AI) has become one of the most transformative technologies in the modern world. Countries across the globe are investing heavily in AI research, infrastructure and applications.

AI allows computer systems to perform tasks that generally require human intelligence, such as:

- Learning from information
- Understanding language
- Recognising images
- Making decisions
- Solving complex problems

AI is becoming an important factor in economic development, national security and technological competition among countries.

Global AI Development

Countries are developing AI capabilities to improve:

- Industrial production
- Healthcare systems
- Defence technology
- Scientific research
- Digital services

Major Areas of AI Application

1. AI in Healthcare

AI is transforming healthcare systems worldwide.

Applications include:

Disease Diagnosis

AI systems analyse medical information to identify diseases at an early stage.

Examples:

- Medical image analysis
- Cancer detection
- Disease prediction

Drug Development

AI helps researchers:

- Identify new medicines

- Study biological processes
- Reduce research time

Personalised Healthcare

AI can analyse patient data and suggest suitable treatments.

2. AI in Education

AI is changing the education sector through:

- Personalised learning systems
- Digital tutors
- Automated evaluation
- Online education platforms

3. AI in Defence and Security

Countries use AI for:

- Surveillance
- Cybersecurity
- Autonomous systems
- Military data analysis

4. AI in Industry

Industries use AI for:

- Automation
- Quality control

- Predictive maintenance
- Supply chain management

Challenges of Artificial Intelligence

Data Privacy

AI requires large amounts of data. Protecting personal information is a major concern.

Ethical Issues

Important concerns include:

- Bias in AI systems
- Responsible use
- Transparency

Employment Impact

Automation may change traditional jobs and increase demand for new technical skills.

Importance of AI

AI is important because it supports:

- Economic growth
- Scientific innovation
- Better governance

- Technological competitiveness

Important Points

- AI is a major global technology trend.
- Healthcare, defence and industries are major AI application areas.
- Data protection and ethical AI development are important challenges.

SECTION 2: SPACE TECHNOLOGY

2. Global Space Exploration and Cooperation

Introduction

Space technology has become an important area of international cooperation and competition.

Countries use space technology for:

- Scientific research
- Communication
- Navigation
- Weather monitoring
- Defence applications

Importance of Space Technology

Communication

Satellites support:

- Television broadcasting
- Internet services
- Global communication

Weather Forecasting

Satellites help monitor:

- Storms
- Climate patterns
- Natural disasters

Navigation

Satellite navigation systems support:

- Transportation
- Defence
- Emergency services

Earth Observation

Space-based observation helps in:

- Agriculture monitoring
- Environmental studies
- Disaster management

Major Space Organisations

NASA

Full Form:

National Aeronautics and Space Administration

Country:

United States

Established:

1958

Headquarters:

Washington, D.C.

Activities

NASA works on:

- Space exploration
- Planetary research
- Earth science
- Human space missions

European Space Agency (ESA)

Introduction

ESA is a group of European countries cooperating in space activities.

Activities

- Satellite development
- Space science
- Earth observation

China National Space Administration (CNSA)

Introduction

China's space agency manages China's space programme.

Major Areas

- Lunar exploration
- Space stations
- Satellite technology

International Space Cooperation

Countries cooperate in areas such as:

- Space research
- Climate monitoring
- Scientific missions

Challenges in Space

Space Debris

Old satellites and spacecraft parts create risks for future missions.

Space Security

Increasing use of satellites creates concerns about:

- Satellite protection
- Space-based conflicts

Important Points

- Space technology supports communication and scientific development.
- Satellites are important for climate monitoring.
- Space cooperation is increasing globally.

SECTION 3: GLOBAL HEALTH

3. Global Health Security and Disease Prevention

Introduction

Global health security focuses on preventing and controlling diseases that can affect populations across countries.

International cooperation is necessary because diseases can spread quickly across borders.

Major Areas of Global Health

1. Disease Surveillance

Countries monitor diseases through:

- Health data systems
- Laboratory networks
- International cooperation

2. Vaccine Development

Vaccines help prevent infectious diseases.

Research focuses on:

- Faster vaccine development
- Better protection
- Global access

3. Healthcare Infrastructure

Strong healthcare systems require:

- Hospitals
- Medical professionals
- Research facilities

Role of World Health Organization (WHO)

Introduction

WHO is the specialised agency of the United Nations responsible for international public health.

Established

1948

Headquarters

Geneva, Switzerland

Functions of WHO

Disease Control

WHO supports countries in:

- Disease prevention
- Emergency response

Health Guidelines

WHO provides recommendations on:

- Public health practices
- Disease management

Research Cooperation

WHO coordinates international health research.

Global Health Challenges

Emerging Diseases

New diseases can create global health risks.

Antimicrobial Resistance

Excessive use of antibiotics can make medicines less effective.

Healthcare Inequality

Many countries face challenges in accessing quality healthcare.

Important Points

- WHO is the main UN health organisation.
- Global health requires international cooperation.
- Disease prevention and healthcare access are major priorities.

SECTION 4: ENVIRONMENT AND CLIMATE

4. Global Climate Change and Environmental Protection

Introduction

Climate change is one of the biggest environmental challenges facing the world.

It is mainly caused by increasing greenhouse gas emissions from:

- Industries

- Transportation
- Energy production
- Deforestation

Major Effects of Climate Change

1. Rising Global Temperatures

Increasing temperatures lead to:

- Heat waves
- Changing climate patterns
- Environmental damage

2. Extreme Weather Events

Climate change increases:

- Floods
- Droughts
- Storms
- Wildfires

3. Impact on Biodiversity

Climate change affects:

- Wildlife
- Forest ecosystems
- Marine life

International Climate Agreements

Paris Agreement

Adopted:

2015

Objective:

To limit global temperature increase and reduce greenhouse gas emissions.

Climate Finance

Meaning

Climate finance refers to financial support provided for:

- Clean energy projects
- Climate adaptation
- Disaster management

Renewable Energy Transition

Countries are increasing investment in:

- Solar energy

- Wind energy
- Green hydrogen
- Electric vehicles

Important Points

- Climate change is a global issue.
- Renewable energy is a major solution.
- Climate finance supports developing countries.

SECTION 5: GLOBAL REPORTS AND INDEXES

5. Human Development Report

Introduction

The Human Development Report is published by the United Nations Development Programme (UNDP).

It measures development beyond economic growth.

Human Development Index (HDI)

HDI evaluates countries based on three major areas:

1. Health

Measured through:

- Life expectancy

2. Education

Measured through:

- Years of schooling

3. Standard of Living

Measured through:

- Income levels

Importance

HDI provides information about:

- Quality of life
- Social development
- Human progress

Important Points

- Published by: UNDP
- Index: Human Development Index
- Focus: Health, education and income

6. World Economic Reports

Introduction

International organisations publish economic reports to analyse global economic conditions.

Important institutions include:

- International Monetary Fund (IMF)
- World Bank
- World Economic Forum

IMF Reports

Focus areas:

- Economic growth
- Inflation
- Financial stability

World Bank Reports

Focus areas:

- Development
- Poverty reduction
- Infrastructure

Importance

Reports help governments understand:

- Economic challenges
- Development trends
- Policy requirements

SECTION 6: INTERNATIONAL AWARDS AND HONOURS

7. Nobel Prize and International Recognition

Introduction

The Nobel Prize is one of the world's most prestigious international awards.

It recognises outstanding contributions in different fields.

Nobel Prize Categories

Physics

Recognises achievements in physical sciences.

Chemistry

Recognises discoveries and innovations in chemistry.

Medicine

Recognises contributions to medical science.

Literature

Recognises outstanding literary work.

Peace

Recognises efforts for peace and international cooperation.

Economic Sciences

Recognises research in economics.

Importance of Nobel Prize

It encourages:

- Scientific research
- Peace efforts
- Human development

SECTION 7: IMPORTANT INTERNATIONAL APPOINTMENTS

8. Leadership Changes in International Organisations

Introduction

Appointments to international organisations influence global decision-making and policy direction.

Important organisations include:

- United Nations
- IMF
- World Bank
- WHO
- WTO

Importance of International Appointments

Leadership affects:

- Global policies
- International cooperation
- Economic decisions
- Development strategies

Major Areas of Leadership

Economic Organisations

Includes:

- IMF leadership
- World Bank leadership

Focus:

- Global finance
- Development policies

Health Organisations

Includes:

- WHO leadership

Focus:

- Global health programmes
- Disease control

United Nations Bodies

Focus:

- Peace
- Sustainable development
- Human rights

SECTION 1: IMPORTANT REPORTS AND INDICES

1. Human Development Report (HDR)

Introduction

The Human Development Report is one of the most important global development reports. It measures the progress of countries not only through economic growth but also through the overall quality of human life.

The report focuses on improving people's capabilities and opportunities.

Published By

United Nations Development Programme (UNDP)

Human Development Index (HDI)

The main index used in the report is the **Human Development Index (HDI)**.

HDI measures development using three major dimensions:

1. Health

Measured through:

- Life expectancy at birth

- Quality of healthcare facilities
- Overall health conditions

2. Education

Measured through:

- Expected years of schooling
- Average years of schooling

3. Standard of Living

Measured through:

- Gross National Income per capita

Importance of HDI

HDI provides a broader understanding of development.

A country may have high economic growth but still face challenges in:

- Education
- Healthcare
- Income equality

Importance for India

India uses development indicators to improve:

- Healthcare systems
- Education policies

- Poverty reduction programmes

Important Points

- Report: Human Development Report
- Released by: UNDP
- Index: Human Development Index
- Parameters: Health, Education, Income

2. Global Innovation Index (GII)

Introduction

The Global Innovation Index measures the innovation capability and performance of countries.

Innovation plays an important role in:

- Economic growth
- Technology development
- Scientific research

Published By

World Intellectual Property Organization (WIPO)

Areas Measured

The index evaluates countries based on:

1. Research and Development

Includes:

- Scientific research
- Investment in innovation
- Research institutions

2. Human Capital

Measures:

- Skilled workforce
- Education
- Knowledge development

3. Infrastructure

Includes:

- Digital infrastructure
- Communication systems
- Technology facilities

4. Business Innovation

Measures:

- Private sector innovation
- Technology adoption

Importance

The index helps countries understand their strengths and weaknesses in innovation.

Importance for India

India's innovation growth is supported by:

- Technology startups
- Digital economy
- Research institutions
- IT sector

Important Points

- Index: Global Innovation Index
- Organisation: WIPO
- Focus: Innovation and technology development

3. Global Hunger Index (GHI)

Introduction

The Global Hunger Index measures hunger and nutrition levels across countries.

It highlights challenges related to:

- Food security
- Child nutrition
- Health conditions

Published By

The report is prepared by international organisations working on hunger and nutrition issues.

Major Indicators

The index considers:

1. Undernourishment

Measures the proportion of population with insufficient food intake.

2. Child Wasting

Measures children with low weight compared to height.

3. Child Stunting

Measures children affected by long-term nutritional deficiency.

4. Child Mortality

Measures deaths among children due to poor nutrition and health conditions.

Importance

The index helps governments design policies for:

- Nutrition improvement
- Food security
- Child welfare

Importance for India

India focuses on:

- Nutrition programmes
- Food distribution
- Maternal and child healthcare

Important Points

- Focus: Hunger and nutrition
- Indicators: Undernourishment, wasting, stunting, child mortality

4. Climate Change Performance Index (CCPI)

Introduction

Climate change has become one of the biggest global challenges.

The Climate Change Performance Index evaluates countries based on their efforts to reduce greenhouse gas emissions and promote climate action.

Published By

The index is released by organisations including:

- Germanwatch
- NewClimate Institute
- Climate Action Network

Parameters

The index evaluates:

1. Greenhouse Gas Emissions

Measures emission reduction efforts.

2. Renewable Energy

Measures progress in clean energy adoption.

3. Energy Use

Evaluates energy efficiency.

4. Climate Policy

Studies government climate actions.

Importance

The index encourages countries to improve climate performance.

Importance for India

India's climate priorities include:

- Renewable energy expansion
- Solar energy development
- Green hydrogen
- Sustainable development

Important Points

- Index: Climate Change Performance Index
- Focus: Climate action and emission reduction

5. Global Peace Index (GPI)

Introduction

The Global Peace Index measures the level of peacefulness among countries.

It studies security conditions, conflicts and militarisation.

Published By

Institute for Economics and Peace (IEP)

Major Indicators

1. Safety and Security

Includes:

- Crime levels
- Internal stability

2. Ongoing Conflicts

Measures:

- Internal conflicts

- International conflicts

3. Militarisation

Studies:

- Military expenditure
- Armed forces involvement

Importance

The index helps understand:

- Global security trends
- Conflict situations
- Peace-building requirements

Important Points

- Published by: Institute for Economics and Peace
- Focus: Peace and security

SECTION 2: IMPORTANT INDIAN REPORTS AND RANKINGS

6. NITI Aayog Reports and Indices

Introduction

NITI Aayog is India's policy think tank responsible for research, policy recommendations and development monitoring.

Established

2015

Replaced

Planning Commission

Major Functions

Policy Development

Provides suggestions on:

- Economic development
- Social policies
- Governance reforms

Monitoring Development

Tracks progress in:

- Health
- Education
- Innovation
- Sustainable development

Cooperative Federalism

Promotes cooperation between:

- Central Government
- State Governments

Important Reports by NITI Aayog

Sustainable Development Goals Index

Measures states' progress towards SDGs.

Innovation Index

Measures innovation performance of states.

Health Index

Evaluates healthcare performance.

Importance

NITI Aayog reports help governments:

- Identify challenges
- Improve policies
- Compare performance

Important Points

- Organisation: NITI Aayog
- Established: 2015
- Replaced: Planning Commission

7. RBI Financial Reports

Introduction

The Reserve Bank of India publishes several reports related to India's banking system and financial stability.

Major RBI Reports

Financial Stability Report

Purpose

Examines:

- Banking sector health
- Financial risks
- Economic stability

Monetary Policy Report

Focus

Studies:

- Inflation trends
- Interest rates
- Economic growth

Importance

RBI reports help understand:

- Banking conditions
- Financial risks
- Economic trends

SECTION 3: IMPORTANT INTERNATIONAL SUMMITS AND CONFERENCES

8. G20 Summit

Introduction

The G20 is one of the most important international economic forums.

It brings together major economies to discuss global challenges.

Full Form

Group of Twenty

Main Areas

Economic Growth

Discussion on:

- Global economy
- Investment
- Trade

Climate Change

Focus on:

- Climate finance
- Renewable energy

Digital Economy

Focus on:

- Digital technology
- Innovation

India's Role

India hosted the G20 Summit in 2023.

Major themes included:

- Global South
- Digital public infrastructure
- Sustainable development

Important Points

- G20 focuses on economic cooperation.
- Includes developed and developing economies.

9. United Nations Climate Conferences

Introduction

UN climate conferences bring countries together to discuss climate change solutions.

Main Objectives

Reduce Emissions

Countries work to reduce greenhouse gases.

Climate Finance

Developing countries seek financial support.

Adaptation

Countries develop strategies to manage climate impacts.

Importance

Climate conferences influence:

- Environmental policies
- Energy strategies
- International agreements

SECTION 4: IMPORTANT INTERNATIONAL AGREEMENTS

10. Paris Climate Agreement

Introduction

The Paris Agreement is one of the most important international agreements on climate change.

Adopted

2015

Main Objective

To limit global temperature rise and reduce greenhouse gas emissions.

Key Features

National Climate Targets

Countries submit their own climate action plans.

Global Cooperation

Countries cooperate in:

- Technology sharing
- Climate finance

Climate Adaptation

Support countries affected by climate change.

Importance for India

India focuses on:

- Renewable energy
- Solar power
- Sustainable development

11. International Solar Alliance (ISA)

Introduction

The International Solar Alliance promotes solar energy cooperation among countries.

Founded By

India and France

Established

2015

Headquarters

Gurugram, India

Objectives

- Promote solar energy
- Encourage investment
- Support clean energy transition

Importance

ISA supports global efforts to:

- Reduce carbon emissions
- Increase renewable energy usage

SECTION 1: AWARDS AND HONOURS

1. International Awards and Recognitions

Introduction

Awards and honours recognise outstanding contributions in different fields such as:

- Science
- Literature
- Sports
- Social service
- Peace
- Technology
- Culture

International awards help recognise individuals and organisations that contribute to human development.

Major International Awards

Nobel Prize

Introduction

The Nobel Prize is one of the world's most prestigious awards.

It was established according to the will of Swedish scientist Alfred Nobel.

Nobel Prize Categories

1. Physics

Awarded for major discoveries and inventions in physics.

2. Chemistry

Recognises important contributions in chemical sciences.

3. Physiology or Medicine

Recognises discoveries that improve human health.

4. Literature

Awarded for outstanding literary achievements.

5. Peace Prize

Recognises efforts towards:

- Peace
- International cooperation
- Human rights

6. Economic Sciences

Recognises research contributions in economics.

Importance of Nobel Prize

The Nobel Prize encourages:

- Scientific research
- Peace-building efforts
- Cultural development
- Human welfare

Important Points

- Founder: Alfred Nobel
- Established: 1901
- Country associated: Sweden

2. Indian Civilian Awards

Introduction

India honours individuals who make significant contributions to the nation through civilian awards.

These awards recognise achievements in:

- Public service
- Science
- Arts
- Literature
- Social work
- Sports

Major Civilian Awards

Bharat Ratna

Introduction

Bharat Ratna is India's highest civilian award.

Given For

Exceptional service and achievement in areas including:

- Public affairs
- Science
- Arts
- Literature

Padma Awards

The Padma Awards are among India's highest civilian honours.

They are divided into:

Padma Vibhushan

For exceptional and distinguished service.

Padma Bhushan

For distinguished service of a high order.

Padma Shri

For distinguished service in various fields.

Importance

Awards encourage excellence and recognise national contributions.

SECTION 2: SPORTS CURRENT AFFAIRS

3. Indian Sports Development

Introduction

Sports have become an important part of India's youth development and international identity.

India continues to improve performance in:

- Cricket
- Athletics
- Badminton
- Wrestling
- Shooting
- Chess

- Hockey

Importance of Sports

Sports contribute to:

- Physical fitness
- Youth development
- National pride
- International recognition

Major Areas of Sports Development

1. Sports Infrastructure

Development includes:

- Stadiums
- Training centres
- Sports academies

2. Talent Development

Programmes focus on:

- Identifying young athletes
- Providing professional training
- Financial support

3. Women in Sports

India is promoting greater participation of women athletes.

Important Sports Organisations

International Olympic Committee (IOC)

Introduction

IOC manages the Olympic Games.

Headquarters

Lausanne, Switzerland

Role

- Organises Olympic Games
- Promotes sports worldwide

International Cricket Council (ICC)

Introduction

ICC is the global governing body for cricket.

Functions

- Organises international cricket tournaments
- Maintains cricket rules

Important Points

- Sports development focuses on infrastructure and talent.
- International competitions improve India's global sports presence.

4. Major Sports Events and Achievements

Introduction

Sports achievements are important current affairs topics because they highlight:

- Athlete performance
- National records
- International success

Areas to Remember

Cricket

Important areas:

- International tournaments
- Player achievements
- Team rankings

Athletics

Important areas:

- World championships
- Records
- International medals

Chess

India has become a major chess power due to:

- Young players
- International achievements

Other Sports

Important areas include:

- Badminton
- Wrestling
- Shooting
- Hockey

SECTION 3: BOOKS AND AUTHORS

5. Importance of Books and Authors in Current Affairs

Introduction

Books released by important personalities are frequently asked in competitive examinations.

Questions usually focus on:

- Book title
- Author
- Subject
- Theme

Types of Important Books

1. Autobiographies

These describe:

- Personal journey
- Professional achievements
- Life experiences

2. Political Books

Discuss:

- Governance
- International relations
- Public policy

3. Science and Technology Books

Explain:

- Scientific discoveries
- Technology developments

4. Historical Books

Cover:

- Important events
- Personalities
- Civilisations

Important Points

When studying books remember:

- Author name
- Field of author
- Main theme
- Historical importance

SECTION 4: IMPORTANT DAYS AND THEMES

6. Important Days in July

Important days are observed to create awareness about social, environmental and international issues.

World Population Day

Date:

11 July

Introduction

World Population Day focuses on population-related challenges and sustainable development.

Importance

It highlights issues such as:

- Population growth
- Healthcare
- Education
- Family planning

World Youth Skills Day

Date:

15 July

Introduction

This day focuses on developing skills among young people.

Importance

Skill development helps:

- Employment generation
- Economic growth
- Career opportunities

International Malala Day

Date:

12 July

Introduction

International Malala Day honours Malala Yousafzai's contribution towards girls' education.

Importance

Promotes:

- Education equality
- Women's empowerment
- Access to learning

Kargil Vijay Diwas

Date:

26 July

Introduction

Kargil Vijay Diwas commemorates India's victory in the Kargil War of 1999.

Importance

The day honours:

- Indian Armed Forces
- Soldiers who sacrificed their lives

SECTION 5: PERSONS IN NEWS

7. Important Personalities in News

Introduction

Persons in news include individuals who gain importance because of:

- Awards
- Appointments
- Achievements
- Scientific discoveries
- Sports success
- Social contributions

Categories of Persons in News

1. Political Leaders

Important because of:

- Policy decisions
- International relations
- Governance

2. Scientists and Researchers

Important due to:

- Scientific discoveries
- Technology development

3. Sports Personalities

Important due to:

- Records
- Championships
- International achievements

4. Social Leaders

Important because of:

- Social reforms
- Public service

SECTION 6: OBITUARIES

8. Important Obituaries

Introduction

Obituaries remember personalities who made important contributions in their fields.

Competitive examinations often ask about:

- Person
- Field
- Contribution
- Achievement

Remembering Personalities

Important details:

- Full name
- Profession
- Major contribution
- Awards received

Examples of Fields Covered

Science

Scientists remembered for:

- Research
- Discoveries
- Innovation

Literature

Writers remembered for:

- Books
- Literary contributions

Sports

Athletes remembered for:

- Achievements

- Records

Arts and Culture

Artists remembered for:

- Music
- Cinema
- Traditional arts